House Price Prediction using Machine Learning

Internship Project • Week 1 Report | Kunal Nigewan

Objective

Built a regression model to predict house prices from property features (area, bedrooms, bathrooms, location-related amenities, etc.) and identified which features most strongly influence price, using the Kaggle Housing Prices dataset.

Procedure

- Loaded Housing.csv with Pandas and explored it (head, shape, info, describe).
- Checked for missing values and duplicate rows.
- Converted categorical columns to numeric using One-Hot Encoding.
- Split data into training and testing sets.
- Trained Linear Regression and Random Forest Regression models.
- Evaluated models using MAE, RMSE, and R^2 Score.
- Visualized price distribution, feature correlations, and prediction accuracy.

Model Results

Model	R^2 Score	Notes
Linear Regression	~0.65	Better fit on this dataset
Random Forest Regression	~0.61	Slightly lower accuracy

Charts & Insights

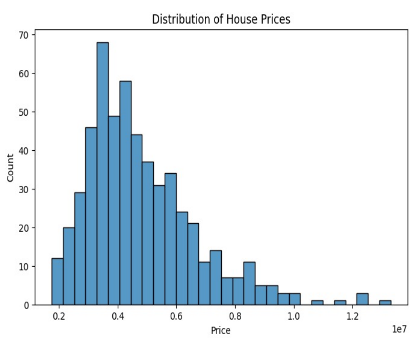

Fig 1: Distribution of House Prices — most homes fall in the medium price range; few outliers are very expensive.

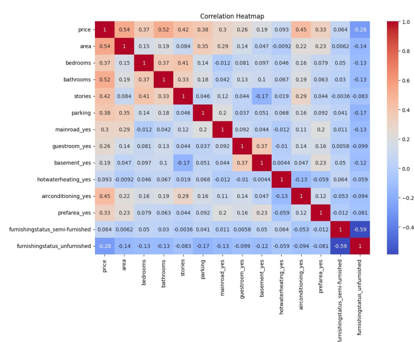

Fig 2: Correlation Heatmap — area, bathrooms, and air conditioning correlate most strongly with price.

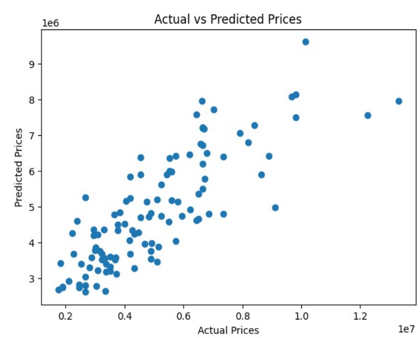

Fig 3: Actual vs Predicted Prices — predictions track actual prices closely, confirming reasonable model fit.

Project Summary

- Loaded, explored, and cleaned the Housing dataset (no missing/duplicate values found).
- Converted categorical features to numeric form via one-hot encoding.
- Trained and evaluated Linear Regression and Random Forest models.
- Linear Regression performed best, with an R^2 score of ~0.65.
- Area, bathrooms, and furnishing status emerged as the strongest price drivers.

Business Insights & Recommendations

- Larger area and more bathrooms consistently correlate with higher prices and demand.
- Furnished homes and those with air conditioning attract more buyer interest.
- Amenities like parking and guest rooms can boost property value.
- Data-driven pricing (via ML models) helps avoid over/under-pricing listings.

Conclusion

This project successfully predicted house prices using machine learning and identified the key factors driving property value. The findings can support real estate businesses in making more accurate, data-backed pricing and marketing decisions.